

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q11-0030

Custer 138 kV  
March 22, 2012

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0030. The Interconnection Customer proposes a 421.7 MW natural gas combined cycle facility interconnecting at the Custer 138 kV substation in the Virginia area. The request was received on April 22, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 26 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,598,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: Tombstone Substation – Sharyland 345.0 kV  
Contingency: Tombstone – Bixby Hill 500.0 kV  
Loading: 107.9% (Rating: 1,152 MVA)  
DFAX: 7.3%  
Pre-project loading: 105.2%
2. Facility: Sideburn – Yavapai 138.0 kV  
Contingency: Rocky Reach – Idaho Springs 138.0 kV  
Loading: 106.8% (Rating: 238 MVA)  
DFAX: 8.6%  
Pre-project loading: 91.6%
3. Facility: Fairfax Substation – Jeune Landing 138.0 kV  
Contingency: Occoquan – Covanta Fairfax Energy 138.0 kV L  
Loading: 104.3% (Rating: 184 MVA)  
DFAX: 29.8%  
Pre-project loading: 36.0%
4. Facility: Sharyland – Bixby Hill 345.0 kV line  
Contingency: Tombstone – Bixby Hill 500.0 kV line  
Loading: 101.9% (Rating: 1,206 MVA)  
DFAX: 7.4%  
Pre-project loading: 99.3%
5. Facility: Loving South – Custer 138.0 kV Line  
Contingency: Fairfax – Franonia 138.0 kV  
Loading: 101.1% (Rating: 242 MVA)  
DFAX: 31.4%  
Pre-project loading: 46.3%

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                    | Required Upgrade           | Cost Estimate | Allocation to Project |
|---------------------------------------------|----------------------------|---------------|-----------------------|
| Tombstone – Sharyland 345.0 kV              | Reconductor line           | \$20,418,000  | 100% (\$20,418,000)   |
| Sideburn – Yavapai 138.0 kV                 | Reconductor line           | \$505,000     | 100% (\$505,000)      |
| Fairfax – Jeune Landing Substation 138.0 kV | Reconductor line           | \$517,000     | 100% (\$517,000)      |
| Sharyland – Bixby Hill 345.0 kV line        | Reconductor line           | \$12,797,000  | 100% (\$12,797,000)   |
| Loving South – Custer 138.0 kV Line         | Reconductor line           | \$361,000     | 100% (\$361,000)      |
|                                             | Total allocated to project |               | \$34,598,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.